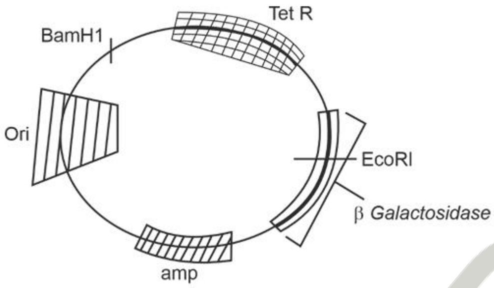
Chapter 9 : Biotechnology : Principles and Processes

1. Polymerase chain reaction (PCR) amplifies DNA following the equation. **[NEET-2025]**

   (1) N^2 (2) 2^n

   (3) $2n + 1$ (4) $2N^2$

2. **[NEET-2025]**



   In the above represented plasmid, an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

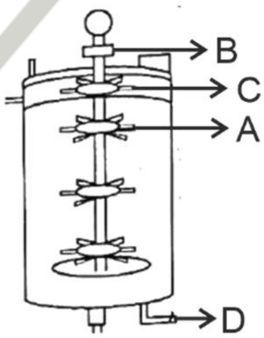
   (1) Using ampicillin & tetracycline containing medium plate.

   (2) Blue color colonies will be selected.

   (3) White color colonies will be selected.

   (4) Blue color colonies grown on ampicillin plates can be selected

3. Identify the part of a bio-reactor which is used as a foam braker from the given figure. **[NEET-2025]**



   (1) A (2) B

   (3) D (4) C

4. Silencing of specific mRNA is possible via RNAi because of **[NEET-2025]**

(1) Complementary dsRNA (2) Inhibitory ssRNA

(3) Complementary tRNA (4) Non-complementary ssRNA

5. The blue and white selectable markers have been developed which differentiate recombinant colonies from non recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate. **[NEET-2025]**

Given below are two statements about this method:

Statement I : The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

Statement II : The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

6. Which of the following enzyme(s) are NOT essential for gene cloning? **[NEET-2025]**

A. Restriction enzymes B. DNA ligase

C. DNA mutase D. DNA recombinase

E. DNA polymerase

Choose the correct answer from the options given below:

(1) C and D only (2) A and B only

(3) D and E only (4) B and C only

7. Given below are two statements : **[NEET-2025]**

Statement I : The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

8. Identify the incorrect statement related to get electrophoresis. **[Re-NEET-2024]**

(1) Separated DNA fragments can be directly seen under UV radiation

(2) Separated DNA can be extracted from get piece

(3) Fragment of DNA moves toward anode

(4) Sieving effect of agarose gel helps in separation of DNA fragments

9. Recombinant DNA molecule can be created normally by cutting the vector DNA and source DNA respectively with: **[Re-NEET-2024]**

(1) Hind II, Hind II (2) Hind II, Alu I

(3) Hind II, EcoR I (4) Hind II, BamHI

10. Select the restriction endonuclease enzymes whose restriction sites are present for the tetracycline resistance (tetR) gene in the pBR322 cloning vector. **[Re- NEET-2024]**

(1) Bam HI and Sal I (2) Sal I and Pst I

(3) Pst I and Pvu I (4) Pvu I and Bam HI

11. Which of the following are correct about EcoR1? **[Re- NEET-2024]**

A. Cut the DNA with blunt end

B. Cut the DNA with sticky end

C. Recognise a specific palindromic sequence

D. Cut the DNA between the base G and A when encounters the DNA sequence 'GAATTC'

E. Exonuclease

Choose the correct answer from the options given below:

(1) B, C, E only (2) A, D, E only

(3) A, C, D only (4) B, C, D only

12. Following are the steps involved in the process of PCR. : **[Re- NEET-2024]**

A. Annealing B. Amplification (~1 billion times)

C. Denaturation D. Treatment with Taq polymerase and deoxynucleotides

E. Extension

Choose the correct sequence of steps of PCR from the options given below :

(1) C → A → D → E → B (2) A → B → E → D → C

(3) A → C → E → D → B (4) D → B → E → C → A

13. What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism? **[NEET-2024]**

A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.

B. It may get integrated into the genome of the recipient.

C. It may multiply and be inherited along with the host DNA.

D. The alien piece of DNA is not an integral part of chromosome.

E. It shows ability to replicate. Choose the correct answer from the options given below:

(1) A and B only
(2) D and E only
(3) B and C only
(4) A and E only

14. Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of: **[NEET-2024]**

(1) 8 bp
(2) 6 bp
(3) 4 bp
(4) 10 bp

15. The "Ti plasmid" of Agrobacterium tumefaciens stands for **[NEET-2024]**

(1) Tumour inhibiting plasmid
(2) Tumor independent plasmid
(3) Tumor inducing plasmid
(4) Temperature independent plasmid

16. Which of the following statements is incorrect? **[NEET-2024]**

(1) A bio-reactor provides optimal growth conditions for achieving the desired product

(2) Most commonly used bio-reactors are of stirring type

(3) Bio-reactors are used to produce small scale bacterial cultures

(4) Bio-reactors have an agitator system, an oxygen delivery system and foam control system

17. Match List I with List II : **[NEET-2024]**

List-I	List-II
A. Lipase	I. Peptide bond
B. Nuclease	II. Ester bond
C. Protease	III. Glycosidic bond
D. Amylase	IV. Phosphodiester bond

Choose the correct answer from the options given below :

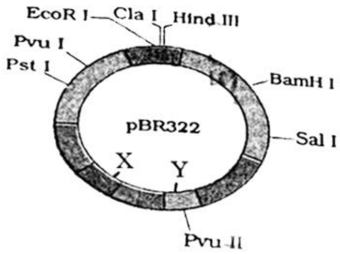
(1) A-IV, B-II, C-III, D-I

(2) A-III, B-II, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-IV, B-I, C-III, D-II

18. The following diagram showing restriction sites in E. coli cloning vector pBR322. Find the role of 'X' and 'Y' genes : **+[NEET-2024]**



(1) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.

(2) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.

(3) The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.

(4) Gene 'X' is responsible for recognitions sites and 'Y' is responsible for antibiotic resistance.

19. Ligation of foreign DNA at which of the following site will result in loss of tetracycline resistance of pBR322? **[NEET-2023-MANIPUR]**

(1) EcoR I (2) BamH I

(3) Pst I (4) Pvu I

20. Thermostable DNA polymerase used in PCR was isolated from : **[NEET-2023-MANIPUR]**

(1) Agrobacterium tumifaciens (2) Bacillus thuringiensis

(3) Thermus aquaticus (4) Escherichia coli

21. Match List-I with List-II. **[NEET-2023-MANIPUR]**

List-I	List-II
(A) Kanamycin	(I) Delivers genes into animal cells
(B) ClaI	(II) Selectable marker
(C) Disarmed retroviruses	(III) Restriction site
(D) Kanamycin Rgene	(IV) Antibiotic resistance

Choose the correct answer from the options given below.

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(2) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

22. Which of the following statement is incorrect about Agrobacterium tumifaciens?

(1) It transforms normal plant cells into tumorous cells. **[NEET-2023-MANIPUR]**

(2) It delivers 'T-DNA' into plant cell.

(3) It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.

(4) 'Ti' plasmid from Agrobacterium tumifaciens used for gene transfer is not pathogenic to plant cells.

23. Which of the following can act as molecular scissors? **[NEET-2023-MANIPUR]**

(1) RNA polymerase (2) DNA polymerase

(3) Restriction enzymes (4) DNA ligase

24. Upon exposure to UV radiation, DNA stained with ethidium bromide will show **[NEET-2023]**

(1) Bright red colour (2) Bright blue colour

(3) Bright yellow colour (4) Bright orange colour

25. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out **[NEET-2023]**

(1) RNA (2) DNA

(3) Histones (4) Polysaccharides

26. In gene gun method used to introduce alien DNA into host cells, microparticles of ________ metal are used. **[NEET-2023]**

(1) Copper (2) Zinc

(3) Tungsten or gold (4) Silver

27. Main steps in formation of Recombinant DNA are given below. Arrange steps in correct sequence.

A. Insertion of recombinant DNA into the host cell **[NEET-2023]**

B. Cutting of DNA at specific location by restriction enzyme

C. Isolation of desired DNA fragment

D. Amplification of gene of interest using PCR

Choose the correct answer from the options given below :

(1) B, C, D, A (2) C, A, B, D

(3) C, B, D, A (4) B, D, A, C

28. Which of the following is not a cloning vector? **[NEET-2023]**

(1) BAC (2) YAC

(3) pBR322 (4) Probe

29. Separation of DNA, fragments is done by a technique known as :- **[Re- NEET-2022]**

(1) Polymerase Chain Reaction (2) Recombinant technology

(3) Southern blotting (4) Gel electrophoresis

30. Match List-I with List-II :- **[Re- NEET-2022]**

List-I	List-II
(a) Gene gun	(i) Replacement of a faulty gene by a normal healthy gene
(b) Gene therapy	(ii) Used for transfer of gene
(c) Gene cloning	(iii) Total DNA in the cells of an organism
(d) Genome	(iv) To obtain indentical copies of a particular DNA molecule

Choose the correct answer from the options given below :-

(1) (a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

(2) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

(3) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

(4) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

31. Pathogenic bacteria gain resistance to antibiotics due to changes in their :- **[Re- NEET-2022]**

(1) Cosmids (2) Plasmids

(3) Nucleus (4) Nucleoid

32. Which of the following methods is not commonly used for introducing foreign DNA into the plant cell? **[Re- NEET-2022]**

(1) Agrobacterium mediated transformation

(2) Gene gun

(3) 'Disarmed pathogen' vectors

(4) Bacteriophages

33. Refer to the following statements for agarose-gel electrophoresis :- **[Re- NEET-2022]**

(a) Agarose is a natural polymer obtained from seaweed.

(b) The separation of DNA molecules in agarose-gel electrophoresis depends on the size of DNA.

(c) The DNA migrates from negatively charged electrode to the positively charged electrode

(d) The DNA migrates from positively charged electrode to the negatively charged electrode. C

Choose the most appropriate answer from the options given below :-

(1) (a) and (b) only (2) (a), (b) and (c) only

(3) (a), (b) and (d) only (4) (b), (c) and (d) only

34. Which one of the following statement is not true regarding gel electrophoresis technique?

(1) The separated DNA fragments are stained by using ethidium bromide. **[NEET-2022]**

(2) The presence of chromogenic substrate gives blue coloured DNA bands on the gel.

(3) Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.

(4) The process of extraction of separated DNA strands from gel is called elution.

35. Given below are two statements :- one is labelled as Assertion (A) and the other is labelled as Reason (R). **[NEET-2022]**

Assertion (A) : Polymerase chain reaction is used in DNA amplification

Reason (R) : The ampicillin resistant gene is used as a selectable marker to check transformation.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

36. In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme ? **[NEET-2022]**

(1) 5' G A A T T C 3'; 3' C T T A A G 5'   (2) 5' C T C A G T 3'; 3' G A G T C A 5'

(3) 5' G T A T T C 3'; 3' C A T A A G 5'   (4) 5' G A T A C T 3'; 3' C T A T G A 5'

37. Given below are two statements:- **[NEET-2022]**

Statement I : Restriction endonucleases recognise specific sequence to cut DNA known as palindromic nucleotide sequence.

Statement II : Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.

In the light of the above statements, choose the most appropriate answer from the options given below:-

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

38. Which of the following is not a desirable feature of a cloning vector ? **[NEET-2022]**

(1) Presence of a marker gene   (2) Presence of single restriction enzyme site

(3) Presence of two or more recognition sites   (4) Presence of origin of replication

39. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out:- **[NEET-2021]**

(1) RNA   (2) DNA

(3) Histones   (4) Polysaccharides

40. Which of the following is a correct sequence of steps in a PCR (Polymerase Chain Reaction) ?

(1) Denaturation, Annealing, Extension   (2) Denaturation, Extension, Annealing **[NEET-2021]**

(3) Extension, Denaturation, Annealing   (4) Annealing, Denaturation, Extension

41. DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as

(1) Yellow bands (2) Bright orange bands **[NEET-2021]**

(3) Dark red bands (4) Bright blue bands

42. When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :- **[NEET-2021]**

(1) Biopiracy (2) Gene therapy

(3) Molecular diagnosis (4) Safety testing

43. Which of the following is not an application of PCR (Polymerase Chain Reaction) ? **[NEET-2021]**

(1) Molecular diagnosis (2) Gene amplification

(3) Purification of isolated protein (4) Detection of gene mutation

44. Plasmid pBR322 has PstI restriction enzyme site within gene ampR that confers ampicillin resistance. If this enzyme is used for inserting a gene for β-galactoside production and the recombinant plasmid is inserted in an E.coli strain **[NEET-2021]**

(1) it will not be able to confer ampicillin resistance to the host cell.

(2) the transformed cells will have the ability to resist ampicillin as well as produce β galactoside.

(3) it will lead to lysis of host cell.

(4) it will be able to produce a novel protein with dual ability.

45. A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :- **[NEET-2021]**

(1) Degenerate primer sequence (2) Okazaki sequences

(3) Palindromic Nucleotide sequences (4) Poly(A) tail sequences

46. During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first? **[NEET-2021]**

(1) Annealing (2) Extension

(3) Denaturation (4) Ligation

47. Match the organism with its use in biotechnology. **[NEET-2020]**

(a) Bacillus (i) Cloning vector thuringiensis

(b) Thermus (ii) Construction of aquaticus first rDNA molecule

(c) Agrobacterium (iii) DNA polymerase tumefaciens

(d) Salmonella (iv) Cry proteins typhimurium

Select the correct option from the following:-

(a) (b) (c) (d)

(1) (iii) (iv) (i) (ii)

(2) (ii) (iv) (iii) (i)

(3) (iv) (iii) (i) (ii)

(4) (iii) (ii) (iv) (i)

48. The specific palindromic sequence which is recognized by EcoR I is :- **[NEET-2020]**

(1) 5' - GGATCC - 3' , 3' - CCTAGG - 5' (2) 5' - GAATTC - 3' , 3' - CTTAAG - 5'

(3) 5' - GGAACC - 3' , 3' - CCTTGG - 5' (4) 5' - CTTAAG - 3', 3' - GAATTC - 5'

49. Choose the correct pair from the following:- **[NEET-2020]**

(1) Exonucleases: - Make cuts at specific positions within DNA

(2) Ligases :- Join the two DNA molecules

(3) Polymerases: Break the DNA into fragments

(4) Nucleases : Separate the two strands of DNA

50. Identify the wrong statement with regard to Restriction Enzymes. **[NEET-2020]**

(1) Sticky ends can be joined by using DNA ligases.

(2) Each restriction enzyme functions by inspecting the length of a DNA sequence.

(3) They cut the strand of DNA at palindromic sites.

(4) They are useful in genetic engineering.

51. The sequence that controls the copy number of the linked DNA in the vector, is termed :-

(1) Recognition site (2) Selectable marker **[NEET-2020]**

(3) Ori site (4) Palindromic sequence

52. First discovered restriction endonuclease that always cuts DNA molecule at a particular point by recognising a specific sequence of six base pairs is:- **[NEET-2020(COVID-19)]**

(1) EcoR1 (2) Adenosine deaminase

(3) Thermostable DNA polymerase (4) Hind II

53. In Recombinant DNA technology antibiotics are used :- **[NEET-2020(COVID-19)]**

(1) to keep medium bacteria-free

(2) to detect alien DNA

(3) to impart disease-resistance to the host plant

(4) as selectable markers

54. Match the following techniques or instruments with their usage :- **[NEET-2020(COVID-19)]**

(a) Bioreactor	(i) Separation of DNA fragments
(b) Electrophoresis	(ii) Production of large quantities of products
(c) PCR	(iii) Detection of pathogen, based on antigen – antibody reaction
(d) ELISA	(iv) Amplification of nucleic acids

(1) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i) (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i) (4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

55. In a mixture, DNA fragments are separated by:- **[NEET-2020(COVID-19)]**

(1) Bioprocess engineering (2) Restriction digestion

(3) Electrophoresis (4) Polymerase chain reaction

56. Spooling is :- **[NEET-2020(COVID-19)]**

(1) Amplification of DNA

(2) Cutting of separated DNA bands from the agarose gel

(3) Transfer of separated DNA fragments to synthetic membranes

(4) Collection of isolated DNA

57. Select the correct statement from the following :- **[NEET-2020(COVID-19)]**

(1) Gel electrophoresis is used for amplification of a DNA segment.

(2) The polymerase enzyme joins the gene of interest and the vector DNA.

(3) Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzymes of optimum conditions.

(4) PCR is used for isolation and separation of gene of interest.

58. Following statements describe the characteristics of the enzyme Restriction endonuclease. Identify the incorrect statement: **[NEET-2019]**

(1) The enzyme cuts DNA molecule at identified position within the DNA

(2) The enzyme binds DNA at specific sites and cuts only one of the two strands.

(3) The enzyme cuts the sugar-phosphate backbone at specific sites on each strand.

(4) The enzyme recognizes a specific palindromic nucleotide sequence in the DNA

59. DNA precipitation out of a mixture of biomolecules can be achieved by treatment with :-

(1) Isopropanol (2) Chilled ethanol **[NEET-2019]**

(3) Methanol at room temperature (4) Chilled chloroform

60. Match the following enzymes with their functions:- **[NEET-2019(Odisha)]**

(a) Restriction (i) Joins the DNA endonuclease fragements

(b) Restriction (ii) Extends primers exonuclease on genomic DNA template

(c) DNA ligase (iii) Cuts DNA at specific position

(d) Taq polymerase (iv) Removes nucleotides from the ends of DNA

Select the correct option from the following: -

(1) a-iii, b-i, c-iv d-ii

(2) a-iii, b-iv, c-i, d-ii

(3) a-iv, b-iii, c-i, d-ii

(4) a-ii, b-iv, c-i, d-iii

61. The two antibiotic resistance genes on vector pBR322 are :- **[NEET-2019(Odisha)]**

(1) Ampicillin and Tetracycline (2) Ampicillin and Chloramphenicol

(3) Chloramphenicol and Tetracycline (4) Tetracycline and Kanamycin

62. A selectable marker is used to: - **[NEET-2019(Odisha)]**

(1) help in eliminating the non-transformants, so that the transformants can be regenerated

(2) identify the gene for a desired trait in an alien organism

(3) select a suitable vector for transformation in a specific crop

(4) mark a gene on a chromosome for isolation using restriction enzyme

63. Given below are four statements pertaining to separation of DNA fragments using gel electrophoresis. Identify the incorrect statements: - **[NEET-2019(Odisha)]**

(a) DNA is negatively charged molecule and so it is loaded on gel towards the Anode terminal

(b) DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.

(c) Smaller the size of DNA fragment larger is the distance it travels through it.

(d) Pure DNA can be visualized directly by exposing UV radiation.

Choose correct answer from the options given below

(1) (a), (c) and (d) (2) (a), (b) and (c)

(3) (b), (c) and (d) (4) (a), (b) and (d)

64. An enzyme catalysing the removal of nucleotides from ends of DNA is:- **[NEET-2019(Odisha)]**

(1) DNA ligase (2) Endonuclease

(3) Exonuclease (4) Protease

65. Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes ? **[NEET-2018]**

(1) Retrovirus (2) Ti plasmid

(3) λ phage (4) pBR322

66. The correct order of steps in Polymerase Chain Reaction (PCR) is :- **[NEET-2018]**

(1) Extension, Denaturation, Annealing

(2) Annealing, Extension, Denaturation

(3) Denaturation, Extension, Annealing

(4) Denaturation, Annealing, Extension

67. The DNA fragments separated on an agarose gel can be visualised after staining with :-

(1) Acetocarmine (2) Aniline blue **[NEET-2017]**

(3) Ethidium bromide (4) Bromophenol blue

68. The process of separation and purification of expressed protein before marketing is called:

(1) Downstream processing (2) Bioprocessing **[NEET-2017]**

(3) Postproduction processing (4) Upstream processing

69. A gene whose expression helps to identify transformed cell is known as :- **[NEET-2017]**

(1) Vector (2) Plasmid

(3) Structural gene (4) Selectable marker

70. Which of the following is not a feature of the plasmids ? **[NEET- I 2016]**

(1) Independent replication (2) Circular structure

(3) Transferable (4) Single - stranded

71. The taq polymerase enzyme is obtained from:- **[NEET- I 2016]**

(1) Thermus aquaticus (2) Thiobacillus ferroxidans

(3) Bacillus subtilis (4) Pseudomonas putida

72. Which of the following is a restriction endonuclease? **[NEET- I 2016]**

(1) Hind II (2) Protease

(3) DNase I (4) RNase

73. Stirred-tank bioreactors have been designed for :- **[NEET- II 2016]**

(1) availability of oxygen throughout the process

(2) ensuring anaerobic conditions in the culture vessel

(3) purification of product

(4) addition of preservatives to the product

74. A foreign DNA and plasmid cut by the same restriction endonuclease can be joined to form a recombinant plasmid using :- **[NEET- II 2016]**

(1) Polymerase-III (2) Ligase

(3) Eco RI (4) Taq polymerase

75. Which of the following is not a component of downstream processing ? **[NEET- II 2016]**

(1) Preservation (2) Expression

(3) Separation (4) Purification

76. Which of the following restriction enzymes produces blunt ends ? **[NEET- II 2016]**

(1) Xho I (2) Hind III

(3) Sal I (4) Eco RV

ANSWER KEY

1. (2)
2. (3)
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